

CSC236 Project Milestone (2): Project View and Database Design of OOP2 Software

Project Group Number (in Blackboard): 6

Section Number: FA2

Project Title: Smart Trips and Events Management



Leader	ID	Member Name
☐	[REDACTED]	[REDACTED]
☐	[REDACTED]	[REDACTED]
☒	[REDACTED]	[REDACTED]
☐	[REDACTED]	[REDACTED]
☐	2240003762	Rawan Ibrahim Alhanfoush
☐	[REDACTED]	[REDACTED]
☐	[REDACTED]	[REDACTED]

Entity-Relationship (ER) Design:

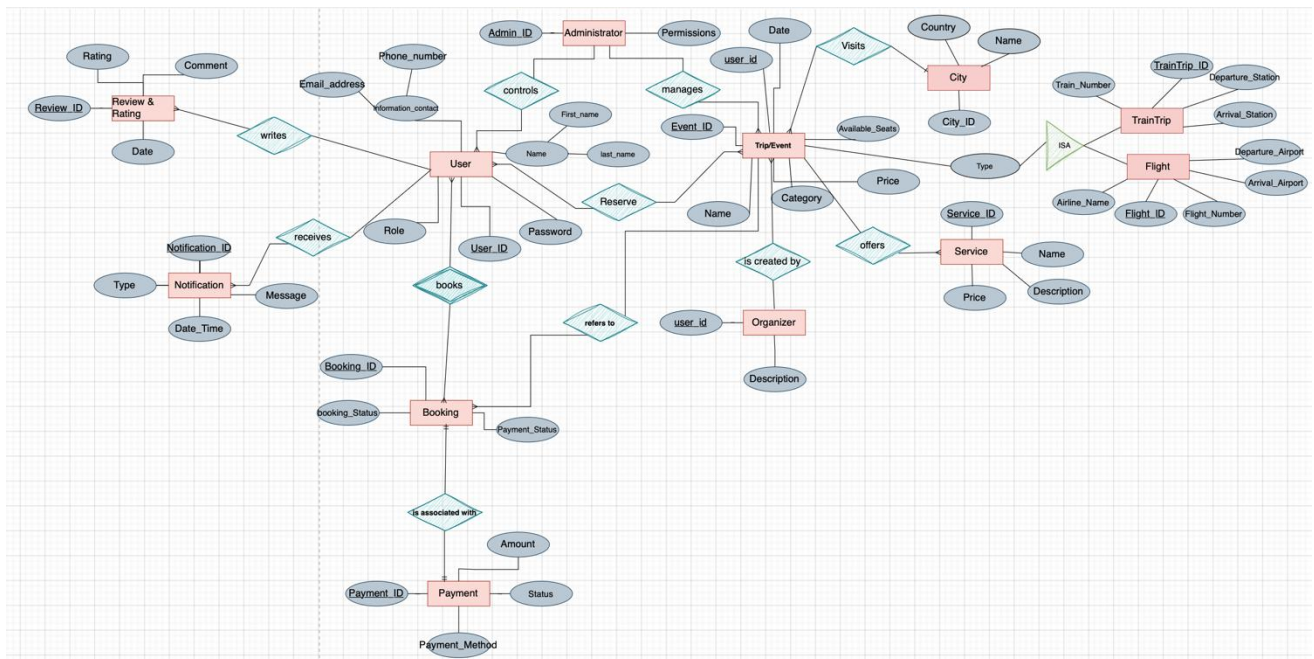


Figure 1: ER Diagram of Trips and Events Management System.

-<https://drive.google.com/file/d/1DIN4WZEBcemjxpV2AeEUIZKr2YAoRlRS/view?usp=sharing>

Database Schema Design:

1- Result of mapping the ER schema diagram into a relational database schema:

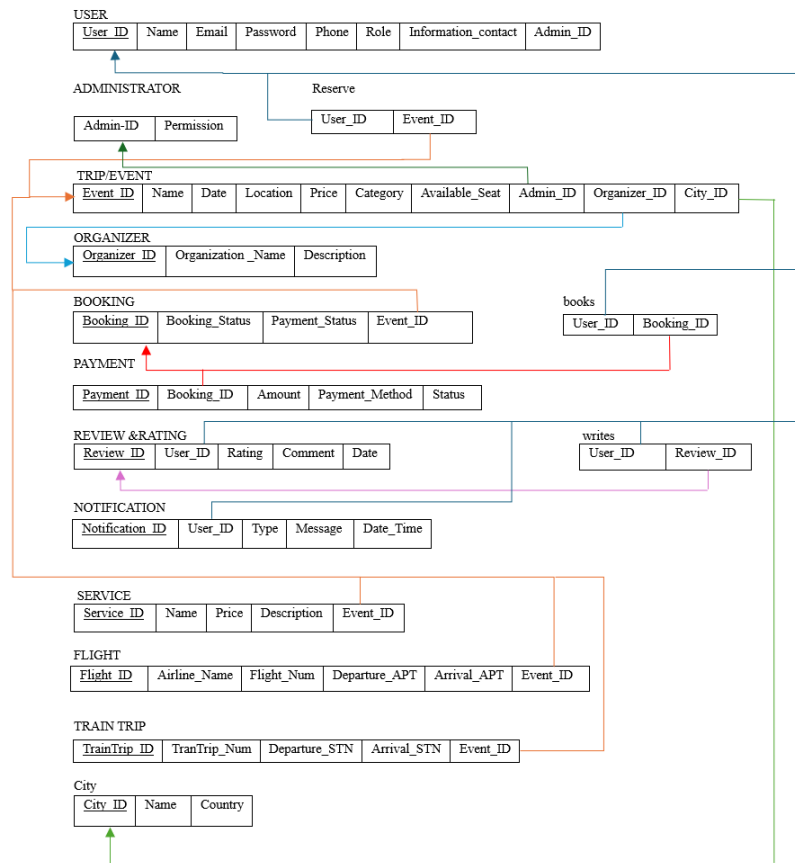


Figure 2: Mapping ER Shema of Trips and Events Management System.

2- Explanation of Relationships:

- **User – Booking (1:M)**: One user can have many bookings.
- **Booking – Payment (1:M)**: One booking can have many payments.
- **User – Notification (1:M)**: One user can receive many notifications.
- **User – ReviewRating (1:M)**: One user can write many reviews.
- **Organizer – TripEvent (1:M)**: One organizer can create many trip events.
- **TripEvent – Flight (1:1)**: A flight is a specific type of trip event.
- **TripEvent – TrainTrip (1:1)**: A train trip is a specific type of trip event.
- **TripEvent – Service (1:M)**: One trip event can offer many services.
- **Admin – User (1:M)**: One administrator can control many users.
- **Admin – TripEvent (1:M)**: One administrator can manage many trip events.
- **TripEvent – City (M:1)**: Many trip events can belong to one city.
- **Booking – TripEvent (M:1)**: Many bookings can be for one trip event.
- **ReviewRating – TripEvent (M:1)**: Many reviews can be written for one trip event.

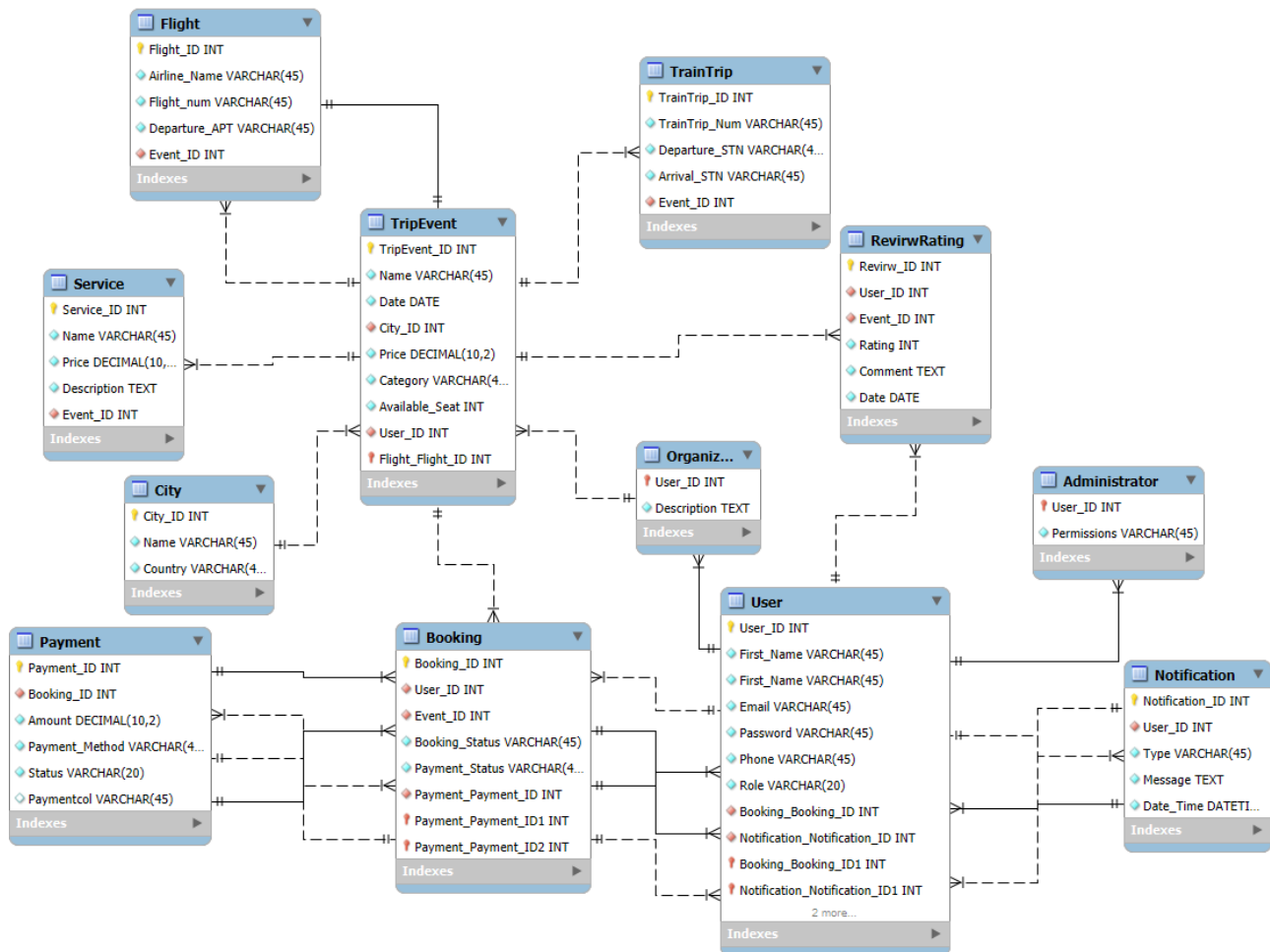


Figure 3: Explanation of The Relationships of Trips and Events Management System.

Database Tables Population:

User Table:

Figure 4 illustrates the User table, which keeps a record of every user in the system, who can be a customer, organizer, or administrator. It holds basic information like first name, surname, email, phone number, and login credentials. There's a role column to specify the kind of user they are.

User_ID	First_Name	Last_Name	Email	Password	Phone	Role
1	Ali	Ahmed	ali@email.com	password123	0501234567	Customer
2	Sara	Khalid	sara@email.com	securepass	0509876543	Organizer
3	Admin	User	admin@email.com	adminpass	0512345678	Admin
4	Mohammed	Salman	mohammed@email.com	pass123	0523456789	Customer
5	Fatima	Hassan	fatima@email.com	securepass	0539876543	Customer
6	Yousef	AlHarbi	yousef@email.com	pass456	0548765432	Customer
7	Layla	Nasser	layla@email.com	mypass	0557654321	Customer
8	Ahmed	Tariq	ahmedtariq@email.com	pass1	0500000001	Organizer
9	Nora	Faisal	nora@email.com	pass2	0500000002	Organizer
10	Omar	Yazid	omaryazid@email.com	pass3	0500000003	Organizer
11	Lina	Saeed	lina@email.com	pass4	0500000004	Organizer
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 4: Data Records of the User Table.

Organizer Table

The Organizer table shown in Figure 5 is only used by users who organize and manage events. It uses the same user ID in the User table and includes a short description of the organizer or work. This isolates information for the organizer but still leaves it attached to their underlying user account.

User_ID	Description
2	Organizes hiking and adventure trips
8	Provides guided city tours
9	Specializes in AlUla tourism
10	Entertainment events in Riyadh
11	Watersports and diving in Jeddah
NULL	NULL

Figure 5: Data Records of the Organizer Table.

TripEvent Table

The TripEvent table shown in Figure 6 holds data for all the events and trips offered in the system. It has the name, date, price, event type, number of seats available, and who is hosting it. Each event also connects to a city through the City table.

Event_ID	Name	Date	City_ID	Price	Category	Available_Seat	User_ID
1	Rock Climbing	2025-08-20	4	250.00	Adventure	15	2
2	Dadan Tour	2025-07-15	4	60.00	Adventure	25	2
3	Maraya Hall Tour	2025-07-22	4	90.00	Cultural	30	9
4	National Museum	2025-06-30	1	10.00	Cultural	50	10
5	Riyadh Season	2025-11-30	1	200.00	Entertainment	40	10
6	Diving Trips	2025-06-29	2	365.00	Adventure	10	11
7	Ithra Visit	2025-07-14	3	35.00	Cultural	20	8
8	Escape Room	2025-09-10	3	260.00	Entertainment	10	8
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 6: Data Records of the TripEvent Table.

Booking Table

The booking table shown in Figure 7 retains data whenever a person reserves a trip. It shows who reserved whom to what event and the booking status up until now and whether payment has been made or not. It helps organizers and users in keeping a tab on who is going to what.

Booking_ID	User_ID	Event_ID	Booking_Status	Payment_Status
1	1	1	Confirmed	Paid
2	1	2	Pending	Unpaid
3	4	1	Confirmed	Paid
NULL	NULL	NULL	NULL	NULL

Figure 7: Data Records of the Booking Table.

Payment Table

This table shown in Figure 8 holds all the payment details of bookings. It shows the amount paid, payment method (e.g., cash or credit card), and if the payment was successful or not. A payment is associated with one booking.

Payment_ID	Booking_ID	Amount	Payment_Method	Status
1	1	250.00	Credit Card	Completed
2	3	250.00	Cash	Pending
NULL	NULL	NULL	NULL	NULL

Figure 8: Data Records of the Payment Table.

ReviewRating Table

They can also leave a review after visiting a trip. Their events they have visited, comments, and rating (out of 5) are stored in ReviewRating table shown in Figure 9. It gives potential customers a sense of expectation and organizes some useful feedback.

Review_ID	User_ID	Event_ID	Rating	Comment	Date
1	1	5	5	Amazing experience!	2025-06-11
2	1	4	4	Very informative tour.	2025-07-16
NULL	NULL	NULL	NULL	NULL	NULL

Figure 9: Data Records of the ReviewRating Table.

Notification Table

The Notification table shown in Figure 10 is utilized to store and send messages to users, like reminders or booking confirmations. Every message is associated with a user and indicates the kind of notification and when it was sent.

Notification_ID	User_ID	Type	Message	Date_Time
1	1	Booking Confirmation	Your trip has been booked.	2025-06-05 10:00:00
2	2	Payment Reminder	Payment for City Tour is pending.	2025-07-10 15:00:00
NULL	NULL	NULL	NULL	NULL

Figure 10: Data Records of the Notification Table.

Service Table

Certain excursions have added services like photography or VIP transport. This table shown in Figure 11 lists those services and relates them to the event they belong to. It includes the name, price, and short description for each.

Service_ID	Name	Price	Description	Event_ID
1	Photography Package	50.00	Professional photography during the event.	1
2	VIP Transport	100.00	Luxury transportation for the tour.	2
NULL	NULL	NULL	NULL	NULL

Figure 11: Data Records of the Service Table.

Flight Table

This table shown in Figure 12 stores flight data when an event entails or implies air travel. It contains the airline, flight number, and departure and arrival terminals. It's linked to the event, so everything is in one location.

Flight_ID	Airline_Name	Flight_Num	Departure_APT	Arrival_APT	Event_ID
1	Saud Airlines	SV101	RUH	JED	2
NULL	NULL	NULL	NULL	NULL	NULL

Figure 12: Data Records of the Flight Table.

TrainTrip Table

The TrainTrip table shown in Figure 13 stores data related to train. It shows the train number, the start and end points, and the event it belongs to. In our project, we kept "SAR" as the train number for every trip.

TrainTrip_ID	TrainTrip_Num	Departure_STN	Arrival_STN	Event_ID
1	SAR	Dammam Station	Riyadh Station	1
2	SAR	Riyadh Station	AlKhobar Station	3
NULL	NULL	NULL	NULL	NULL

Figure 13: Data Records of the TrainTrip Table.

Administrator Table

Admins are users who have the power to administer and manage the system. This table shown in Figure 14 recalls each admin's ID as well as a brief description of his permissions. It distinguishes the admin-level user from standard users.

User_ID	Permissions
3	Full Access
NULL	NULL

Figure 14: Data Records of the Administrator Table.

City Table

The table City shown in Figure 15 includes all the cities in which an event is permitted to occur. Each city has its ID and recalls the name and the country. The events use this table so that we can put them based on place.

City_ID	Name	Country
1	Riyadh	Saudi Arabia
2	Jeddah	Saudi Arabia
3	AlKhobar	Saudi Arabia
4	AlUla	Saudi Arabia
NULL	NULL	NULL

Figure 15: Data Records of the City Table.

Application Structure (forms' organization):

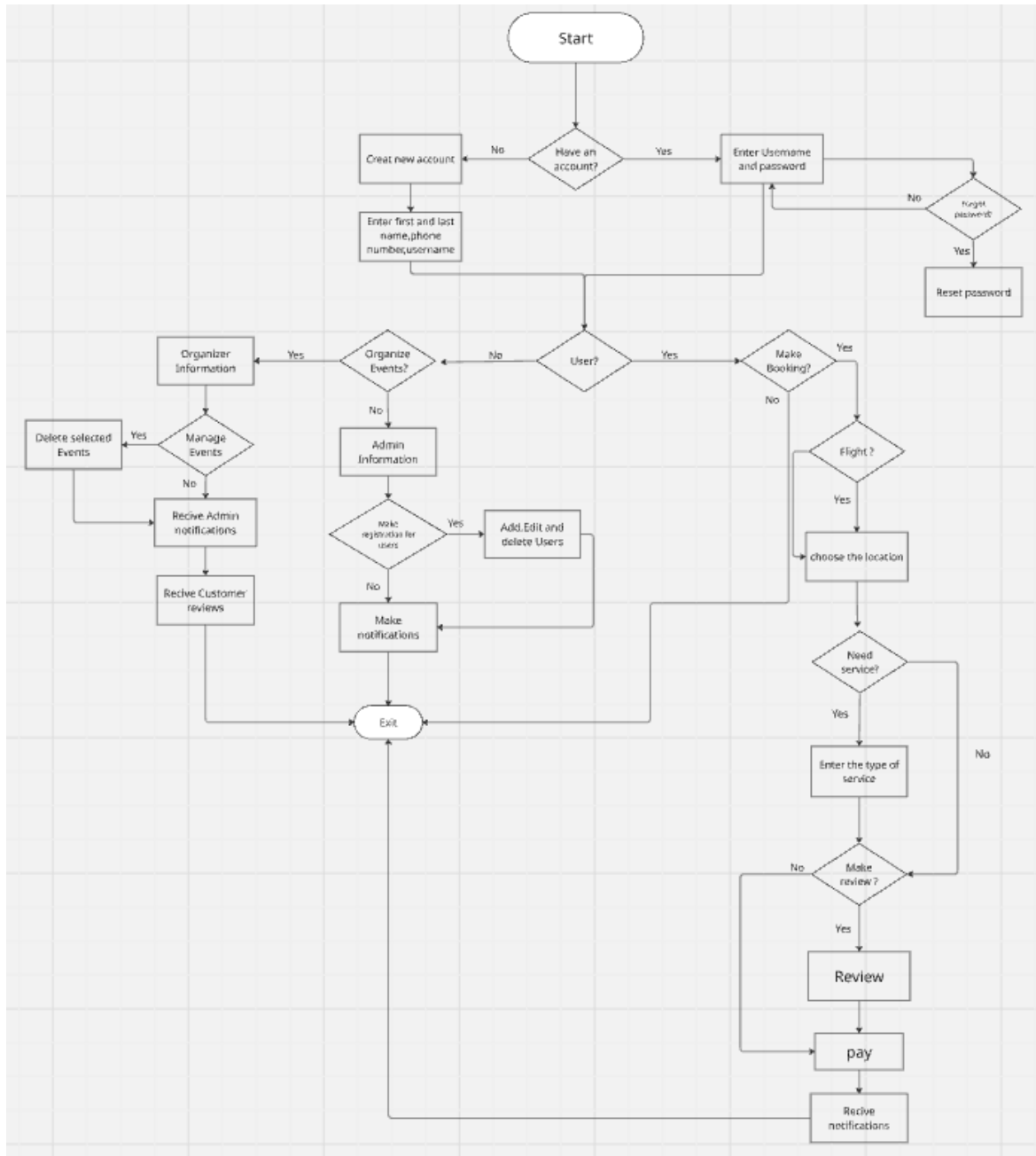


Figure 16: Flowchart Showing the Application Process.

[Flowchart - Miro](#)



Application Interface Design:

Log In:

Figure 17: Login Screen of the Application.

It will display and require users:

- Username: Each user has his unique username.
- Password: to make sure the accounts are saved.
- Create one: for the user who enters our application for the first time.

Create Account:

Figure 18: Create Account Screen for New Users.



After the user clicks the create one button, he will be redirected to this interface:

It contains 6 fields to collect the user's credentials first name, last name, phone, username, password, confirm password. After that he uses the combo box and chooses his purpose of using our program as a customer, admin, or organizer.

Customer interface:

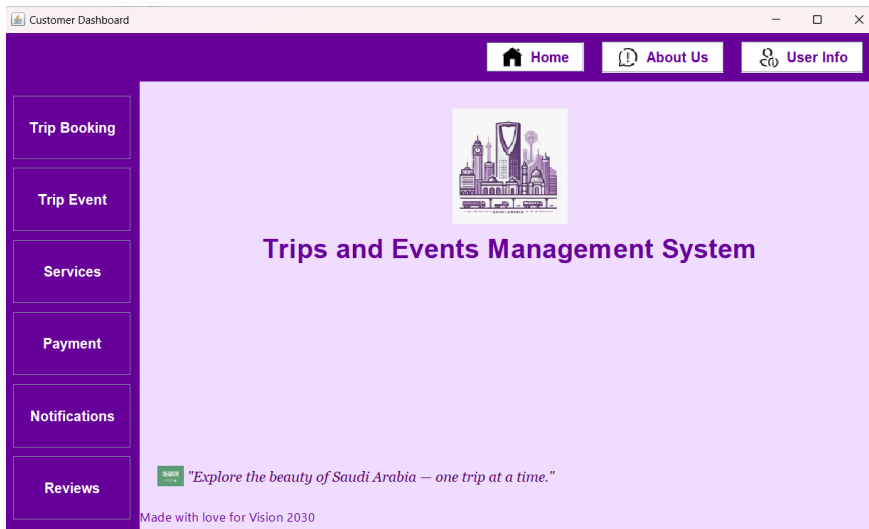


Figure 19: Customer Home Screen After Login.

Customer interface will be displayed after customer log-in as customer by entering his username and password.

Customer functionality:

- Trip Booking: The customer chooses the city and how to transport.
- Trip Event: The customer chooses the city and event he wants to experience.
- Services: The customer chooses if he wants additional special services.
- Payment: The customer chooses the payment method.
- Notification: User alerts arrive from the admin.
- Reviews: The user writes his opinion about the application and events.
- Home: Welcome home page.
- About us: Program Information and Contact Information.
- User info: Customer's contact information.

About Us:

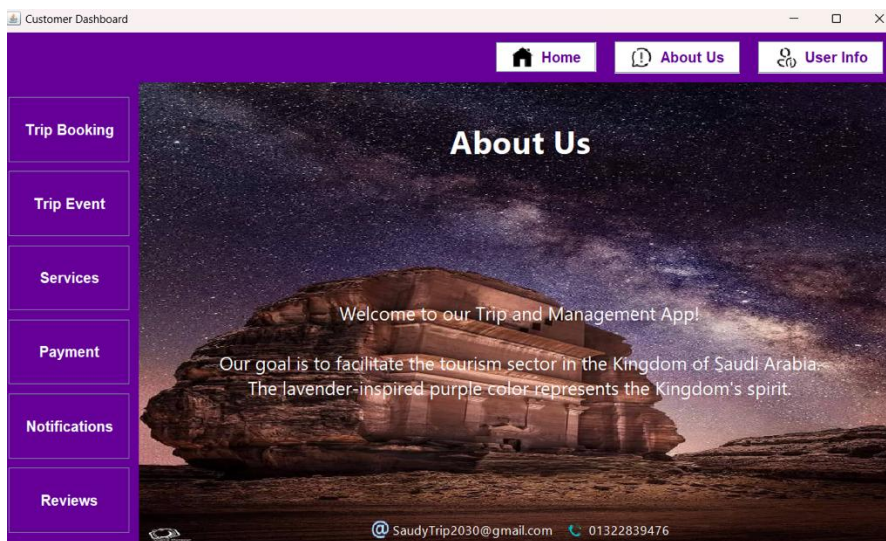


Figure 20: About Us Page of the Application.

Introductory text about the objective of our application and our contact information.

User Information:

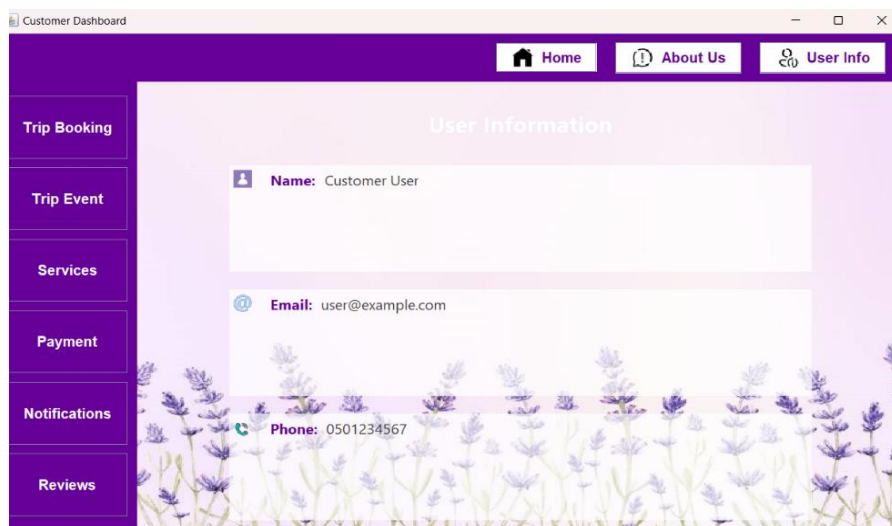


Figure 21: User Information Screen.

Displays the customer's contact information that he has registered.

Trip Booking:

Figure 22: Trip Booking Interface.

The customer chooses one of the modes of transport, plane or train, and then chooses the city in which he is and the city to which he will travel and chooses the date of travel.

Trip Event:

Figure 23: Trip Event Selection Page.

The customer selects the same city he wants to travel to and then chooses the date and time of the event he will go to.

Serves:

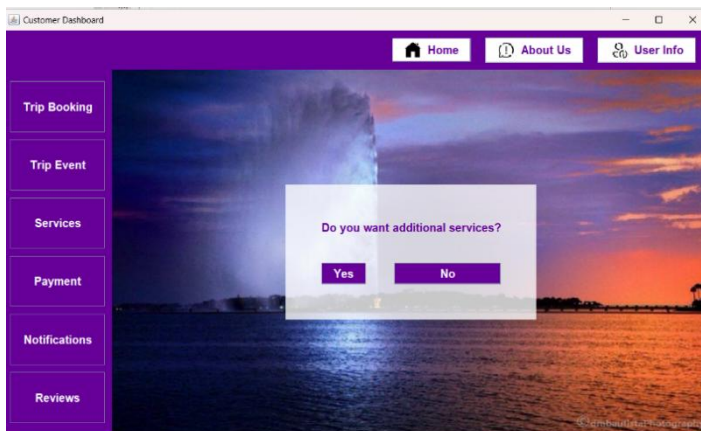


Figure 24: Service Selection Prompt.

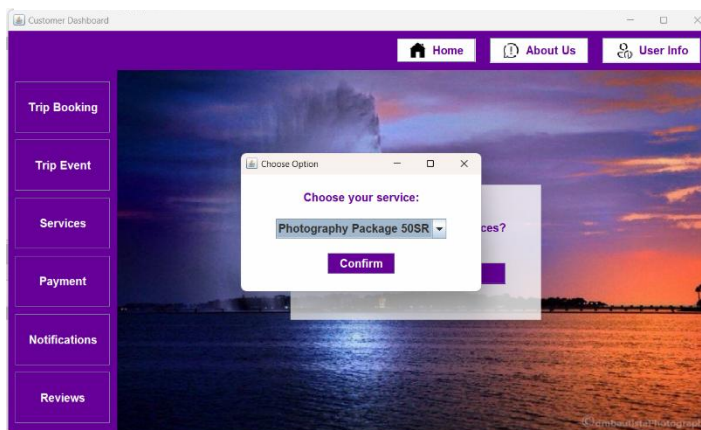


Figure 25: Available Services Dropdown.

The customer chooses if he wants additional services or not, if he chooses the option Yes, a page appears to him with a combo box about the available services.

Payment:

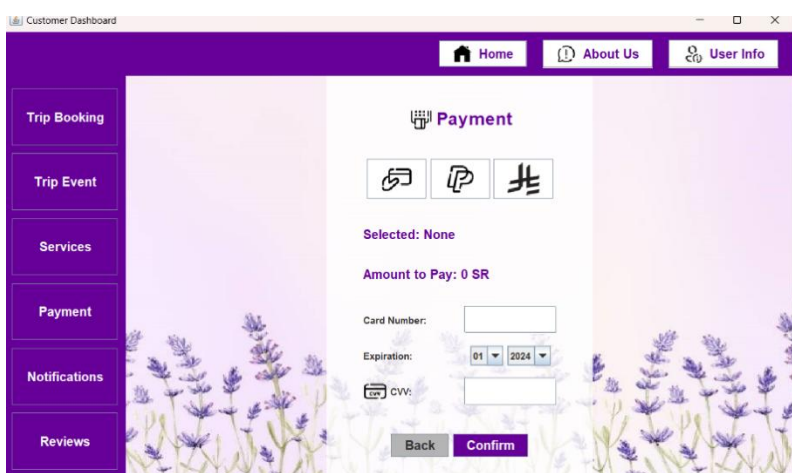


Figure 26: Payment Method Selection Screen.

The customer chooses the payment method that suits him, a credit card, PayPal or cash, then if he chooses a credit card or PayPal, he will fill up the fields required of him.

Notification:

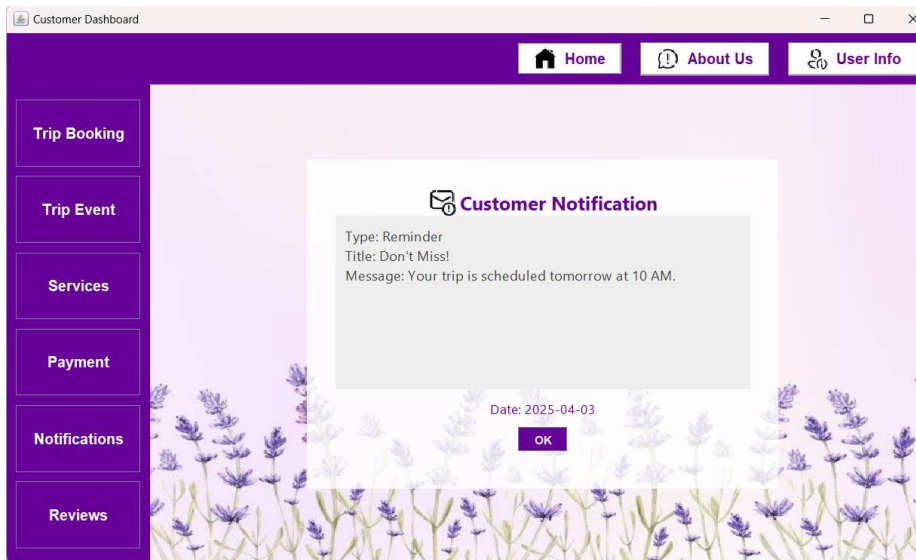


Figure 27: Notification Screen.

Notifications sent by the admin to the customer are displayed with today's date.

Reviews:

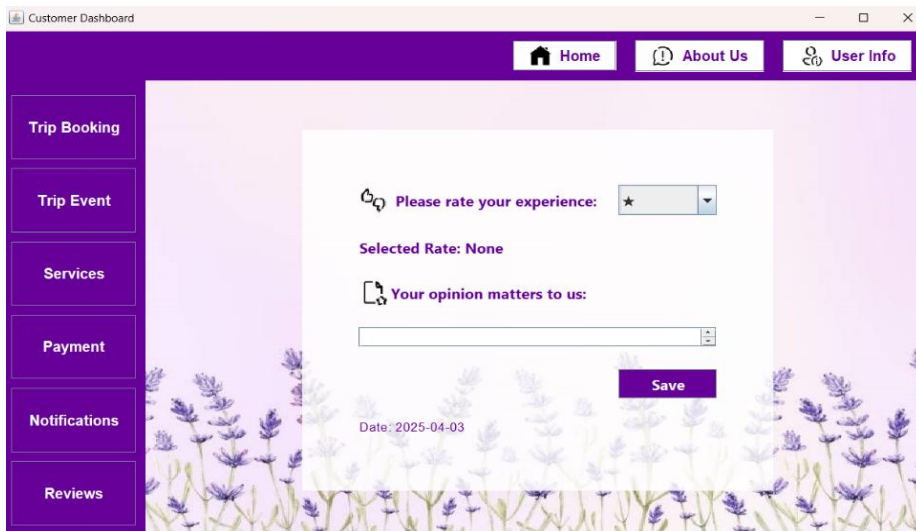


Figure 28: Review Submission Screen.

The customer evaluates the experience of the full application of five stars and writes his opinion on the text area.

Admin interface:

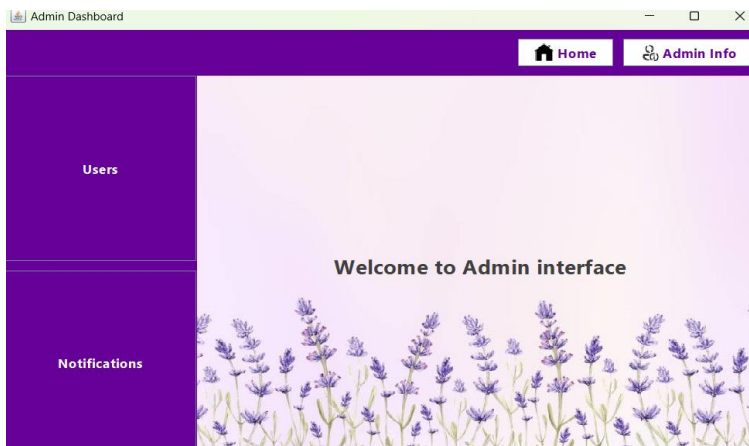


Figure 29: Admin home screen displayed after successful login.

Admin interface will be displayed after admin log-in as a customer by entering his username and password.

Admin functionality:

- Users: Shows user data from organizers and customers.
- Notifications: Sends alerts to customers and organizers.
- Home: Welcome message for the admin.
- Admin info: View the admin contact information.

Admin info:

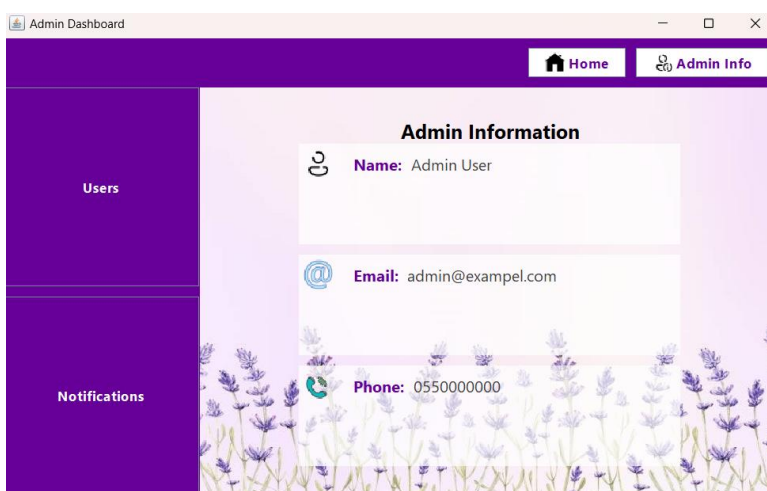


Figure 30: Admin information page showing contact details.

View Admin contact information.

Users:

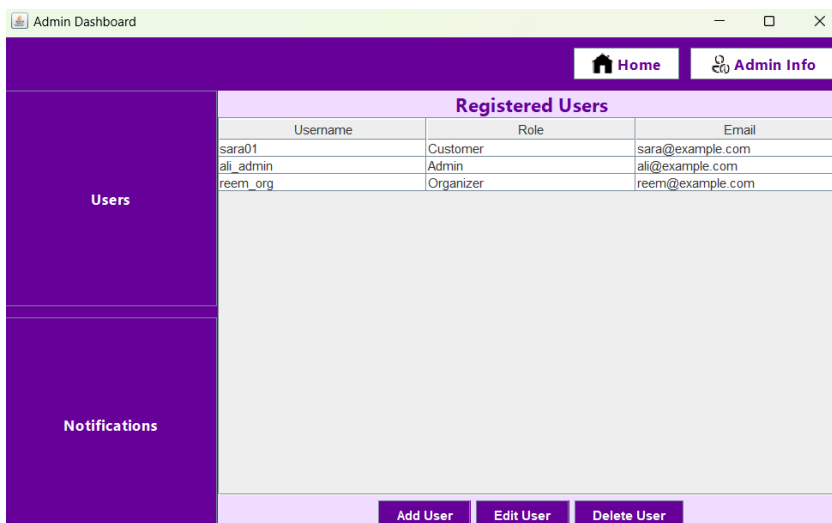


Figure 31: Admin panel showing the list of registered users.

Displays to the admin the private records of users such as administrators, organizers and customers with the possibility of modification, deletion and addition.

Sending Notification:

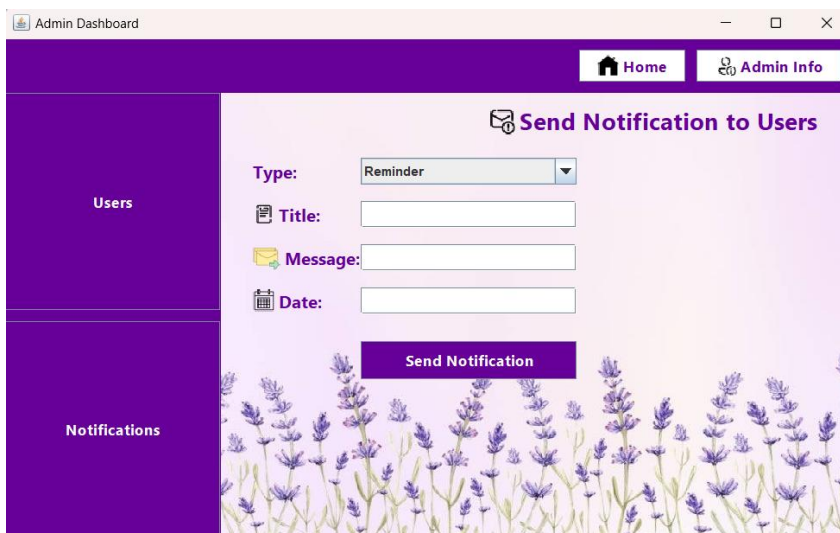


Figure 32: Admin screen for sending notifications to users.

The admin sends messages to users with the choice of the type of message and writes a message in text area with writing the date in which the message was written.

Organizer interface:

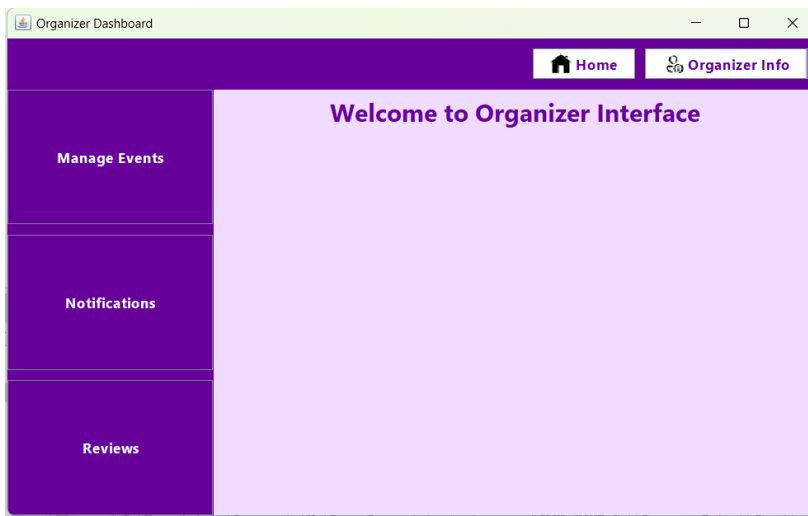


Figure 33: Organizer main interface after login.

Organizer interface will be displayed after organizer log-in as customer by entering his username and password.

Organizer functionality:

- Management Events: The organizer can delete events.
- Notifications: Receiving messages from the Admin.
- Reviews: Receiving feedback from customers.
- Home: Welcome message for the Organizer.
- Organizer info: View the contact information of the organizer.

Organizer Info:

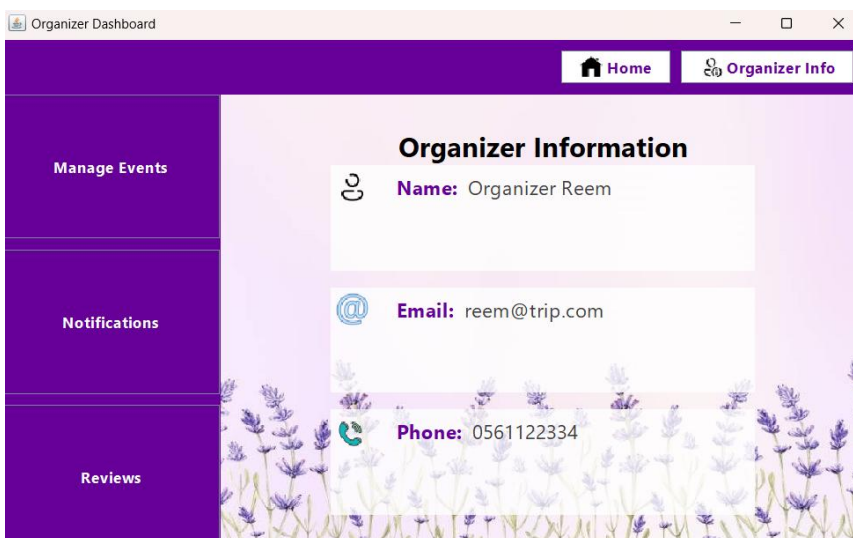


Figure 34: Organizer contact information screen.

View the contact information of the organizer

Manganate Events:

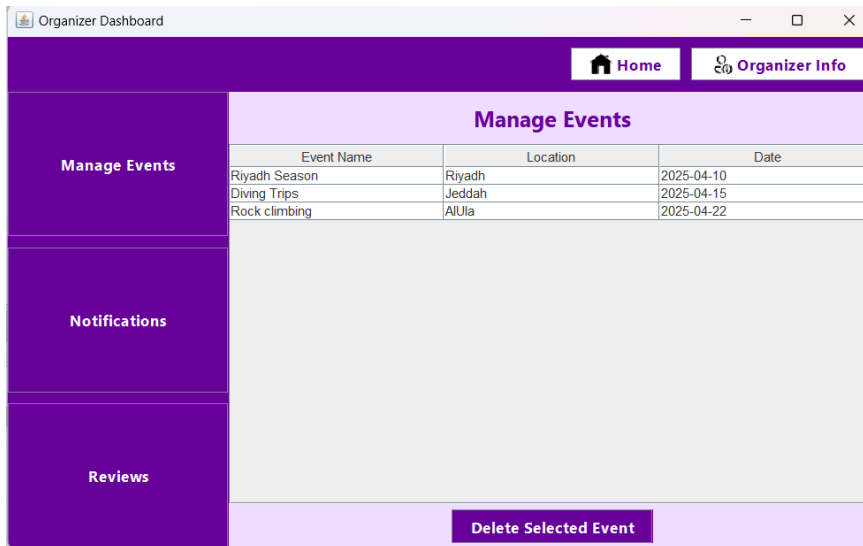


Figure 35: Organizer event management screen.

Organizers can see and delete event records

Receiving Notifications:

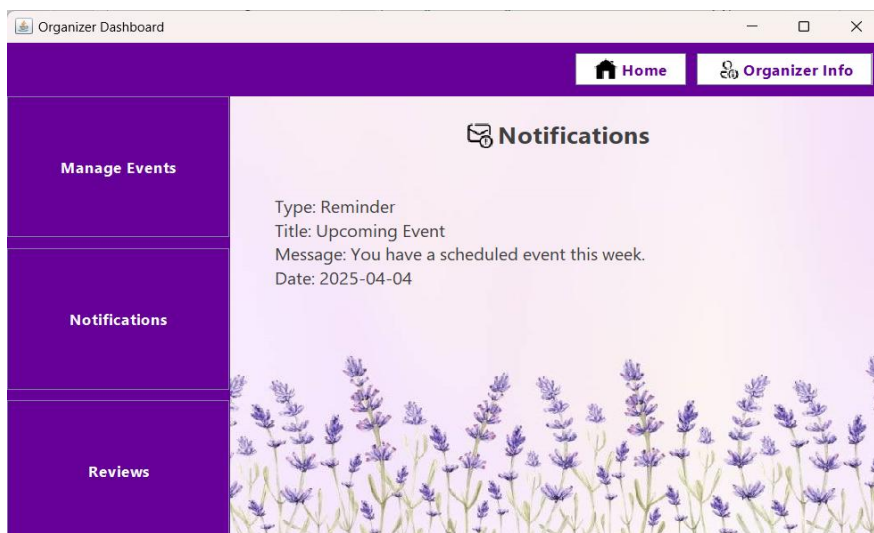


Figure 36: Organizer notification view from admin.

Receiving the details of the message sent by the admin with the date of sending the message.

Reviews:

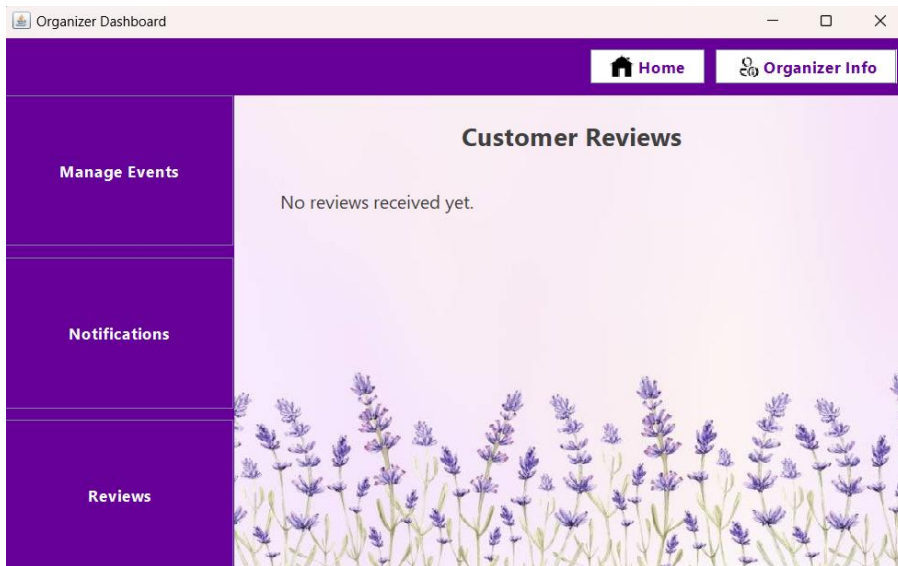


Figure 37: Organizer review screen showing customer feedback.

The organizers receive feedback from customers about the event and the experience they have had.

